DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech **Branch:** Electronics and Computer Engineering/ Electronics and Computer Science **Semester:** V

Subject Code & Name: (BTECPC502) & **(**Digital Signal & Image Processing)

Max Marks: 60 Date: 08/02/2025 **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**		12
1	The value of continuous impulse function δ(t) at $t \neq 0$	L2 / CO1	1
	a. 0 b. 1 c. ∞ d. None		
2	All periodic signals are:	L2 / CO1	1
	a. Power Signals b. Energy Signals c. Neither energy nor power d. None		
3	The system is said to be linear if it follows the principles:	L2 / CO1	1
	a. Homogeneity b. Superposition c. Both a & b d. None		
4	Identify no. of multiplications to be performed in order to calculate N-point DFT:	L3 / CO2	1
	a. N^2 b. $(N^2\text{-}N)$ c. Log N d. N *log(N)		
5	ROC of a Z-transform of discrete unit step causal signal is represented as:	L3 / CO2	1
	a. |z|>1 b. |z|<1 c. 1 <|z|<2 d. |z|>2		
6	Identify the morphological operation performed on image a. Dilation b. Erosion c. Boundary extraction d. All of the above mentioned	L3 / CO5	1
7	Identify the true statement related to Convolution and Correlation operations on a given image: a. Image is pre-rotated by 180^0 for correlation b. Image is pre-rotated by 180^0 for convolution c. Image is pre-rotated by 90^0 for correlation d. Image is pre-rotated by 90^0 for convolution	L3 / CO4	1
8	Histogram Matching is also referred as:	L2 / CO4	1
	a. Histogram equalisation b. Histogram specification		

	c. Histogram linearisation d. None of the mentioned		
9	Piecewise Linear Transformation function involves which of the following? a. Bit-plane slicing b. Intensity level slicing c. Contrast stretching d. All of the mentioned	L2 / CO3	1
10	Segmentation is a process of a. Low level b. Edge level c. Mid-level d. High level	L2 / CO5	1
11	A Spatial Low pass filter whose all coefficients are equal is referred: a. Median Filter b. Box or Mean Filter c. Max Filter d. Min Filter	L2 / CO4	1
12	The step which extracts image components useful in the representation and description of shape of object referred as: a. Segmentation b. Representation & description c. Compression d. Morphological processing	L2 / CO3	1
Q. 2	**Solve the following.**		12
A)	Compare Analog Systems with Digital systems in terms of Bandwidth limitation, System design, Accuracy, Storage and processing of low frequency signal.	L3 / CO1	6
B)	Identify whether following signals are periodic or non-periodic. Specify its fundamental period (if signal is periodic in nature) 1. $x(n) = e^{j6\pi n}$ 2. $x(n) = \cos\left(\frac{2\pi n}{3}\right)$	L3 / CO1	6
Q.3	**Solve the following.**		12
A)	Illustrate the following properties of Z-transform with valid justification. 1. Linearity Property 2. Time Shifting Property 3. Differentiation Property	L2 / CO2	6
B)	Find the output of the Discrete system if the system is having the input $x(n) = \{4, 4, 4\}$ and system's impulse response $h(n) = \{2, 2\}$. Use N-point DFT & IDFT.	L2 / CO2	6
Q. 4	**Solve Any Two of the following.**		12
A)	Explain the various fundamental steps in Digital Image Processing in details	L2 / CO3	6
B)	Find whteher the mentioned two pixels P & Q in a given Image (I) are: 1. 4-Adjacent 2. 8-Adjacent 3. m-Adjacent	L2 / CO3	6

Assume: V={1,2,3}

5	**6**	**8**	**4**	**3(Q)**
4	**9**	**1**	**2(P)**	**4**
5	**2**	**6**	**2**	**1**

C) Identify the Euclidean Distance (D_e), City-block Distance (Dc) and Chess-board Distance (D_b) between the pixels P & Q of a given Grayscale Image $I(x, y)$ — L3 / CO3 — 6

5	**9**	**8**	**7**	**6**
0	**1(P)**	**6**	**2**	**4**
6	**8**	**7**	**2**	**4**
3	**3**	**2**	**6**	**5(Q)**

Specify the result as (D_e, D_c, D_b)

Q.5 Solve Any Two of the following. — 12

A) Apply the grey level transformation on the given image having the intensity range (42 – 115) into transformed image having the intensity range (162 – 224). — L2 / CO4 — 6

$$F(x, y) = \begin{bmatrix} 45 & 87 & 65 & 75 \\ 48 & 109 & 114 & 89 \\ 93 & 68 & 98 & 100 \end{bmatrix}$$

Specify the result of operation.

B) Consider a 3-bit image (L=8) of size 8 x 8 pixels with the histogram specified in the given table. Perform Histogram Equalization. — L3 / CO4 — 6

r_k	0	1	2	3	4	5	6	7
n_k	8	10	10	2	12	16	4	2

Specify the result of operation.

C) Explain the concept of Spatial filtering and illustrate the following Spatial Low pass filters with suitable examples: — L2 / CO4 — 6

1.Mean Filters 2.Weighted average Filters 3. Median Filters

Q. 6 Solve Any Two of the following. — 12

A) Apply the following Morphological operations on the given image. — L3 / CO5 — 6

1. Opening
2. Closing

Image = $\begin{bmatrix} 0111 \\ 1111 \\ 0111 \\ 1111 \end{bmatrix}$ & Structuring Element = $\begin{bmatrix} 11 \\ 11 \end{bmatrix}$

Specify the result of operations.

B) Explain Splitting and Merging image segmentation algorithms in detail. — L2 / CO5 — 6

C) Illustrate the following morphological operations with suitable example — L2 / CO5 — 6

1. Hit or miss transformation
2. Boundary Extraction

***** End *****